

PI-Grau (Internet Protocols)

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- **Topic 5: Corporate Networks revisited: VPN**

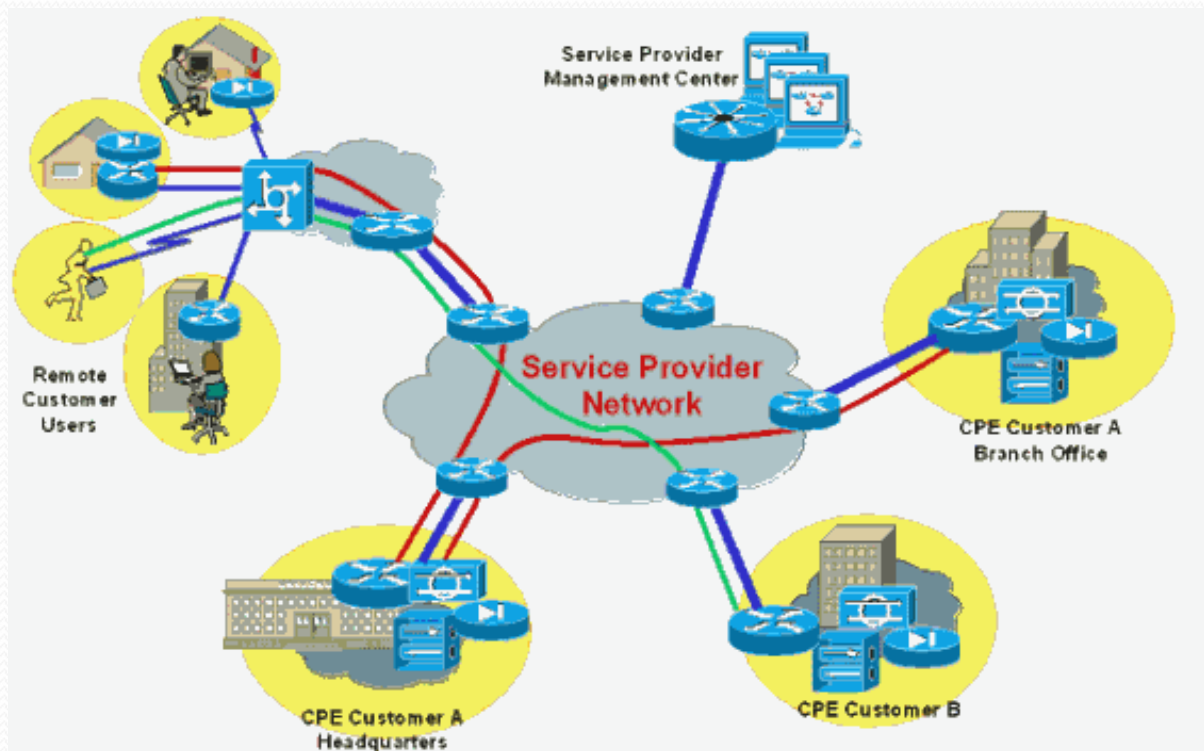
- Objectives

- Connectivity with **the Remote Sites and between sites**
- **Virtual Private Networks (VPN)**
 - MPLS-BGP
 - Metro-Ethernet

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- **VPN (Virtual Private Networks):**

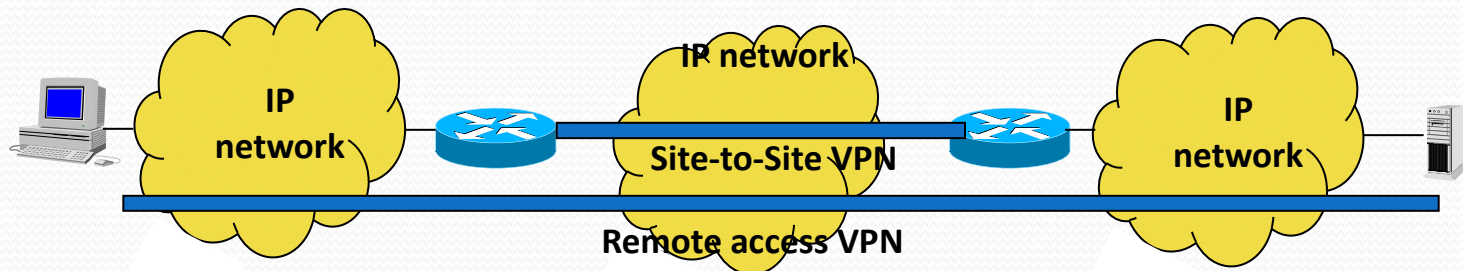
- Network provided by ISP's (L3) or Telecom operators (L2) that interconnects the main site with remote sites or with end users (tele-working) in a secure and private manner.



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● VPN (Virtual Private Networks):

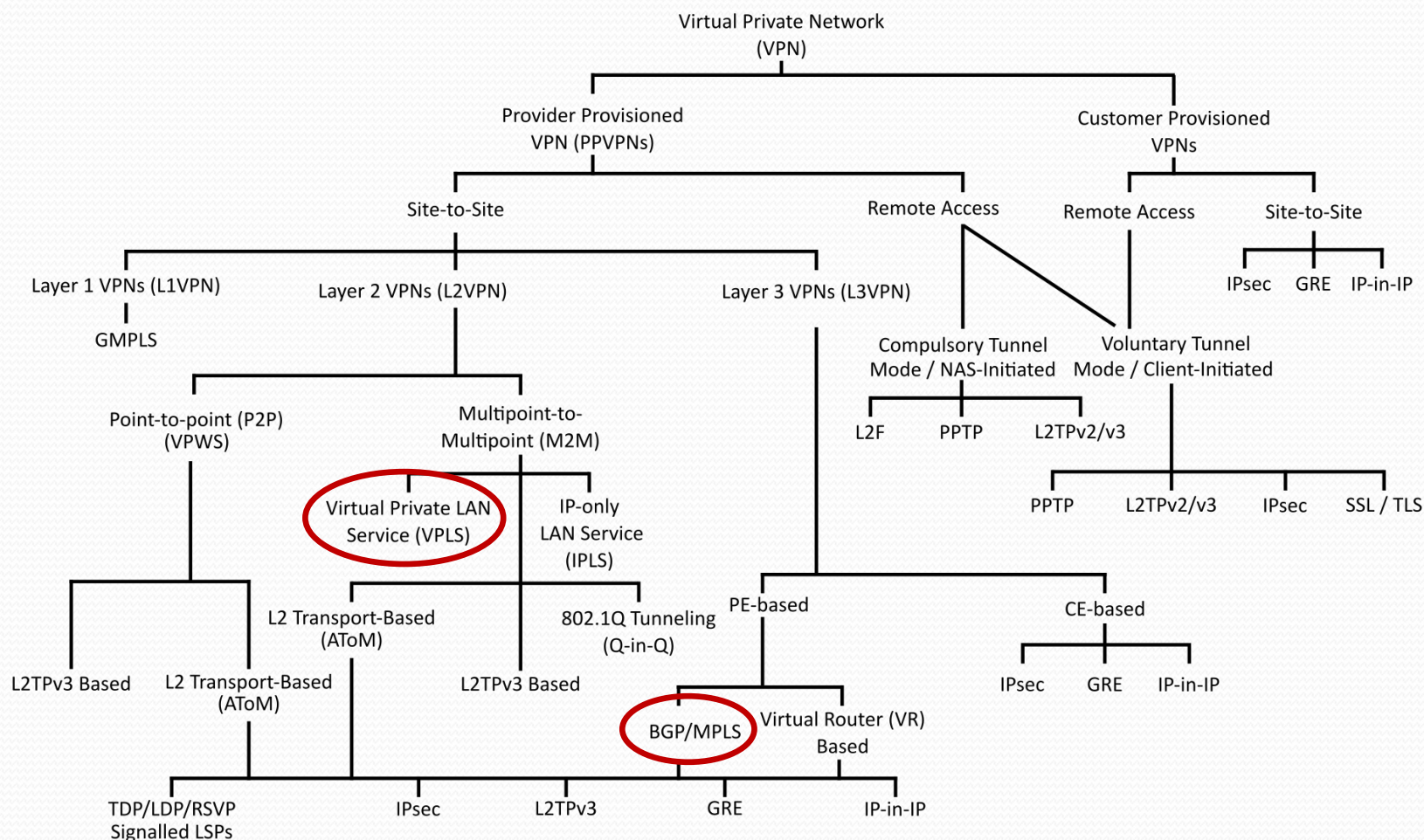
- Connection between two network points using tunnel techniques that may include QoS negotiation, security, ...
- Used to connect the Main Site with the Remote Site
 - **Site-to-Site VPN:** connects two networks (sites), thus implying gateway-to-gateway tunnels. The main example is a remote office connection to the main company quarters.
 - **Remote access VPN:** connects a user to a network, thus implying a host-to-gateway or host-to-host tunnel. The main example is a worker that remotely access to the office.



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• VPN (Virtual Private Networks):

- **Tunnels:** Logic interface that allows packet encapsulation in several formats (L4, L3 or L2). Examples of L2/L3 tunnels are PPTP, L2TP, IP-in-IP Encapsulation, Minimal Encapsulation, Generic Routing Encapsulation, IPsec, MPLS, ...



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● Provider-provisioned VPN (PPVPN):

- PPVPN are L3 tunnels (sometimes are L2) that provide connectivity site-to-site (including many-to-many sites) guaranteeing QoS and security (authentication and encryption) across an ISP,
- They allow to create a super-imposed network topology over the ISP, meaning that the owner of the sites can create a:
 - **Point-to-point topology:** connectivity between two sites,
 - **Point-to-multipoint topology:** connectivity between one site and many sites,
 - **Multipoint-to-multipoint topology:** connectivity between many sites and many sites.
- Examples:
 - **MPLS-BGPv4** that creates tunnels using BGPv4 to signal and create the tunnel and MPLS to route traffic,
 - **VPLS (Virtual Private LAN Service)** that creates VPN's using VLAN (and thus is a emulating a L2 VVPN). The example is **MetroEthernet**.
- In general, VPN's does not support broadcasting services, but there are variants (normally based in VPLS technologies) that overcome this limitation.

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● Traffic Parameters

- **CIR (Committed Information Rate)**

- Average data rate (b/s) associated to a service or flow (not an instantaneous data rate)

- **EIR (Excess Information Rate)**

- Average data rate (b/s) in excess with respect CIR → be careful, sometimes it is specified a PIR (**Peak Information Rate**) and $PIR = CIR + EIR \rightarrow EIR = PIR - CIR$ (an excess !!!)

- **CBS (Committed Burst Size)**

- Size in Bytes of the transmitted information. Amount of bytes that can be sent over a period of time T when congestion occurs. Normally is the **packet size**.

- **EBS (Excess Burst Size)**

- Excess in size in Bytes of the transmitted information. Amount of extra bytes that can sent by a router over the time T when no congestion occurs. Then, if $EBS > 0 \rightarrow$ you can send traffic exceeding the CIR.

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- **Traffic Parameters**

- Example: Capacity of 10 Mb/s with a CIR of 5 Mb/s and a CBS of 1 packet of 1500 Bytes, with an EIR=0 and EBS=0:



The same, but with an EIR=0 and EBS=1



The same, but with an EIR=0 and EBS=2



- Capacity of 10 Mb/s with a CIR of 5 Mb/s and a CBS of 1 packet of 1500 Bytes, with a EBS=1 and an EIR = 1 Mb/s (1 packet of every 10):



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● Traffic Parameters: examples

- **Maximum CIR $\rightarrow \text{CIR} \leq C$ (Line capacity)**
 - The CIR can not exceed the capacity line (It is logical !!!)
- **Best-effort $\rightarrow \text{CIR}=0, \text{CBS}=0, \text{EBS} > 0$**
 - Means that you want a best-effort contract, that means that the ISP does whatever he can do to transfer your data but it is not assuring you that the traffic is sent $\rightarrow$ ideal for a low cost contract
- **CIR enforcement: $\text{rate} \leq \text{CIR}$**
 - The ISP enforces that the traffic is never higher than the CIR
- **CIR Contract: $\text{CIR} > 0, \text{EIR} > 0, \text{CBS} > 0, \text{EBS} > 0$**
 - The customer is guaranteed at least a CIR (under congestion conditions) if he sends his CBS. However, if there is no congestion he can achieved CIR+EIR sending EBS bytes of excess during periods of time.

• Performance or QoS parameters

• Packet Delay

- Delay in seconds of a packet from the time it leaves from a point to the time it arrives to other point
- Important in real-time applications (e.g., VoIP, multimedia) with QoS

• Jitter

- Delay variation of a packet
- Important in real-time applications (e.g., VoIP, multimedia) with QoS
- Normally it is calculated as the difference between the average delay and the minimum delay of packets

• Packet Losses

- 1 less the ratio of delivered packets with respect transmitted packets
- Important for applications such as VoIP (3% of lost packets results in an unacceptable quality reception value)

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● **MPLS (Multi-Protocol Label Switching)**

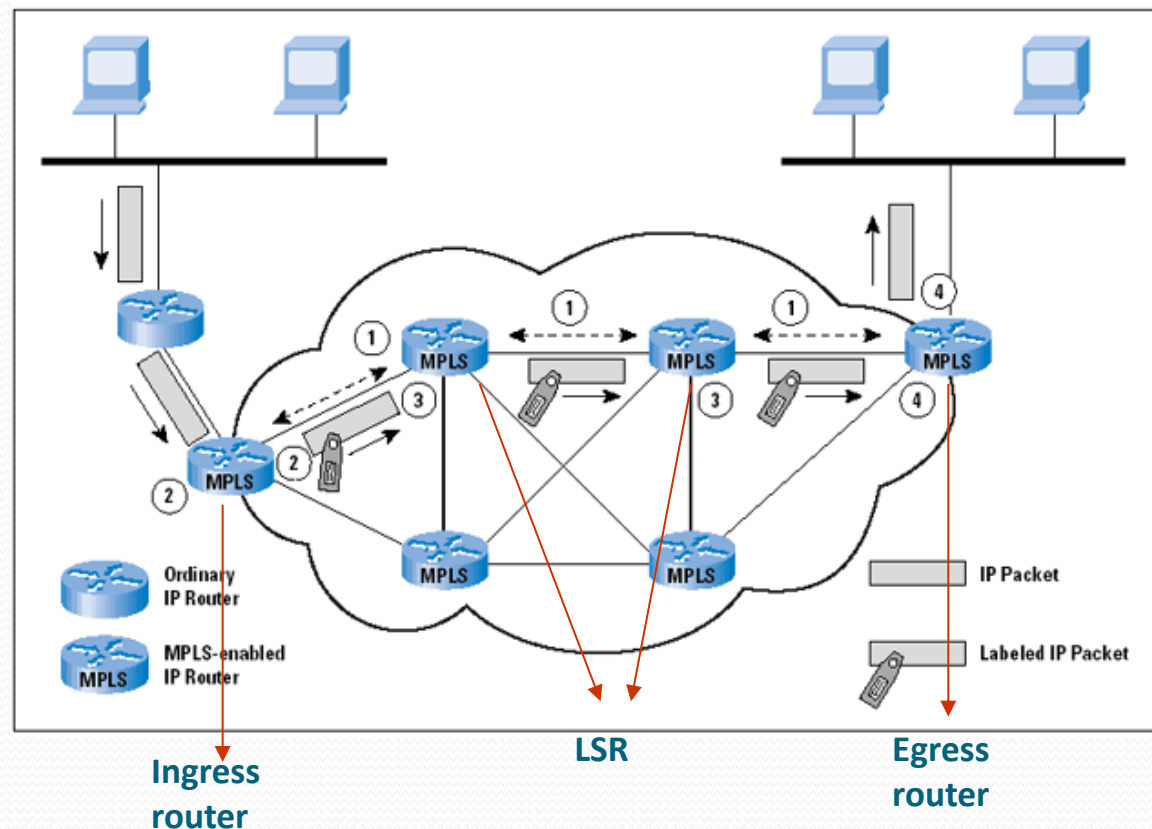
- L3 protocol that forwards packets using labels instead of IP addresses and that allows:
 - **QoS**
 - Guarantee certain capacity to real-time services and guarantee certain packet delay, jitter control and packet losses values → voice and video
 - **VPN services**
 - Support packet segregation in Internet
 - Used jointly with BGP (MPLS-BGP as a VPN)
 - **Traffic Engineering**
 - Optimize network resources from user demand traffic parameters (CIR, EIR, CBS, EBS)
 - **Multi-Protocol support**
 - Independent of the technology (e.g. Frame Relay, ATM, ...)

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• MPLS (Multi-Protocol Label Switching)

- L3 Basic Functionality: packet labeling and packet labeled switching

Figure 1: MPLS Operation



- **LSR (Label Switched Routers)**: routers that switch packets based on labels carried by packets and that identify flows between end points
- **FEC (Forwarding Equivalence Class)**: describes a set of flows that will receive the same treatment (same label) and that correspond to an LSP
- **LSP (Label Switched Path)**: a path in a MPLS network
- **LDP (Label Distribution Protocol)**: protocol for label distribution

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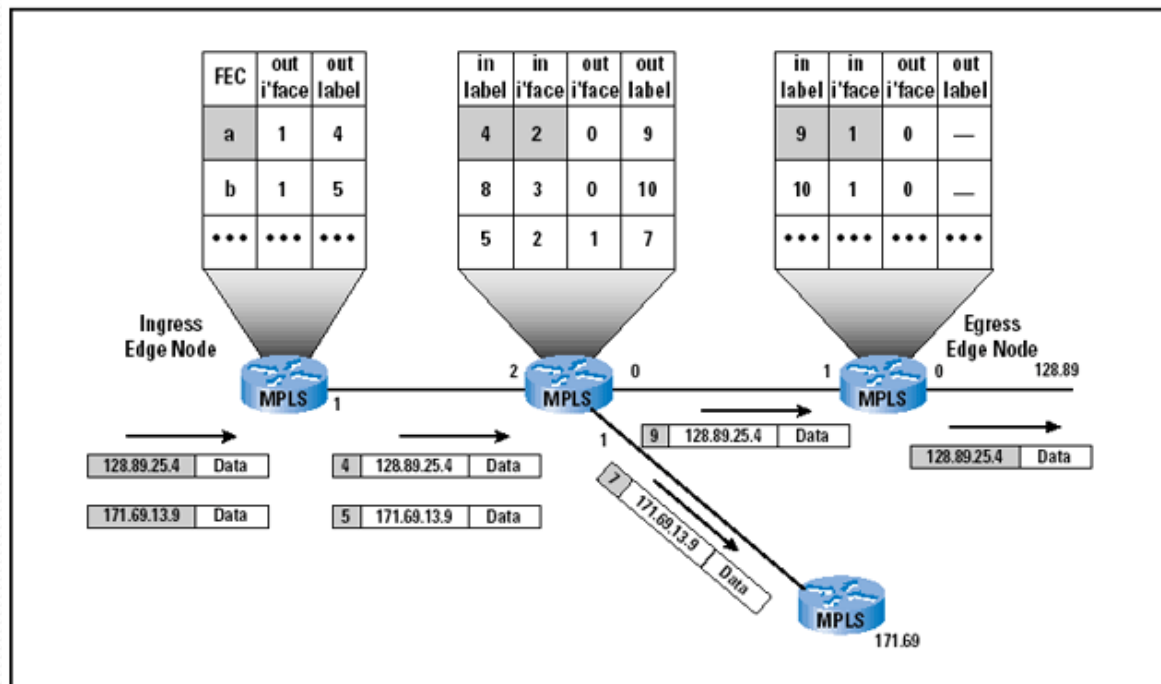
- **LDP (Label Distribution Protocol):** protocol for label distribution
 - A bi-directional communication is established between two LSR routers to exchange labeling information
 - Labels:
 - **Label-Value:** 20-bit to number labels
 - **Exp:** 3-bit to define QoS
 - **S:** 1-bit de flag (bottom of stack) → yes or no
 - **TTL:** 8-bit
 - The label has local meaning, it is to say, a router that receives a packet labeled, checks it, assigns a new one and switches to the output interface



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- Routers route using LSP (Label Switched Path), a pre-established path (e.g., using some IGP routing such as OSPF)
- Packets with different label (belong to a different FEC) will receive a different treatment in the routers (e.g. priorities, schedulers, ...)

Figure 2: MPLS Packet Forwarding

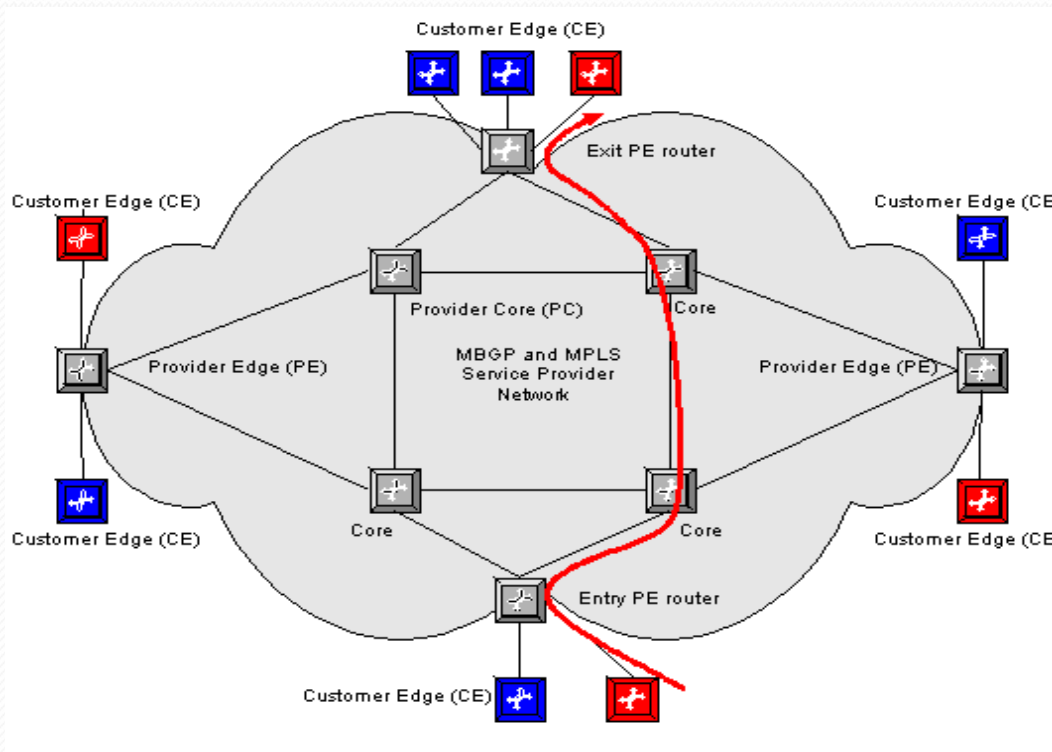


- **Label stacks:** it is allowed to stack labels forming a sequence, so traffic with a same behavior is aggregated (FEC)

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● MPLS-VPN:

- Combines MPLS and BGP to create an IP (L3) VPN (Virtual Private Network)



- **CE:** Customer Edge equipment
- **PE:** Provider Edge equipment
- **P/PC:** Provider (Core) router

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● MPLS-VPN:

- **VPN-IPv4 addresses (12B):** address that identifies the VPN and is composed by a “**Route Distinguisher - RD**” (8 bytes) and an IP@ network → 8B of RD + 4B of @IP = 12B of identifier (e.g. **VPN 146:10.1.1.0** is different of **VPN 37:10.1.1.0**)
- **VPN-IPv4@ should be globally unique** → RD should be globally unique.

RD= 2B (Type Field) + 2B (Admin Field) + 4B (Assigned # Field)

- **Type 0:** The administrator field must contain an AS number (using private AS numbers is discouraged). The assigned field given by the ISP.

RD= 2B (Type Field) + 4B (Admin Field) + 2B (Assigned # Field)

- **Type 1:** The administrator field must contain an IP address (using private IP address space is discouraged). The assigned field given by the ISP.
- **Type 2:** The administrator field must contain a 4-octet AS number (using private AS numbers is discouraged). The assigned field given by the ISP

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- **CE: Customer Edge equipment**
 - Router that give access to the provider
 - Uses E-BGP to announce/learn routes
- **PE: Provider Edge equipment**
 - Exchange routes via BGP with CE
 - Maintains a Virtual Routing and Forwarding (**VRF**) table for each of the connected sites
 - **VRF** defines the VPN membership of a customer site attached to a PE router
 - Exchange VRF information with other PE using **MBGP**
 - **MBGP (Multi-protocol BGP):** BGP version that carries VPN-IPv4 addresses traversing an ISP and that allows to a PE router learn routes from other PE router
 - Acts as ingress and egress point for the Label Switched Paths (LSP): inserts and removes MPLS labels

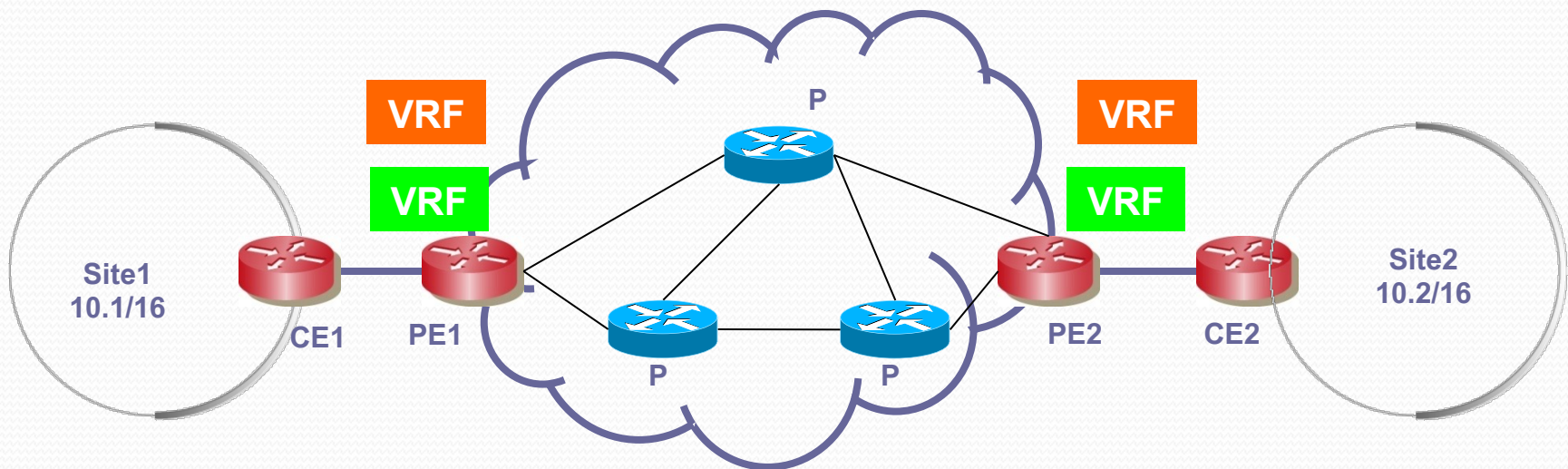
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- **P/PC: Provider (Core) router**
 - Any router that is not attached to a CE
 - Forward traffic between PE routers acting as Label-Switched Routers (LSRs)
 - Only need to know routing information to reach PE routers
 - Required to support LDP at a minimum
- **How to exchange packets between two CE ?**
 - Use BGP to export routes
 - Use **Extended Communities** (8-byte) to filter and associate BGP traffic to a VRF (Virtual Router and Forwarding)
 - VRF are tables associated to PE routers
 - Use MPLS to switch traffic in the Internet core

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● MPLS-VPN:

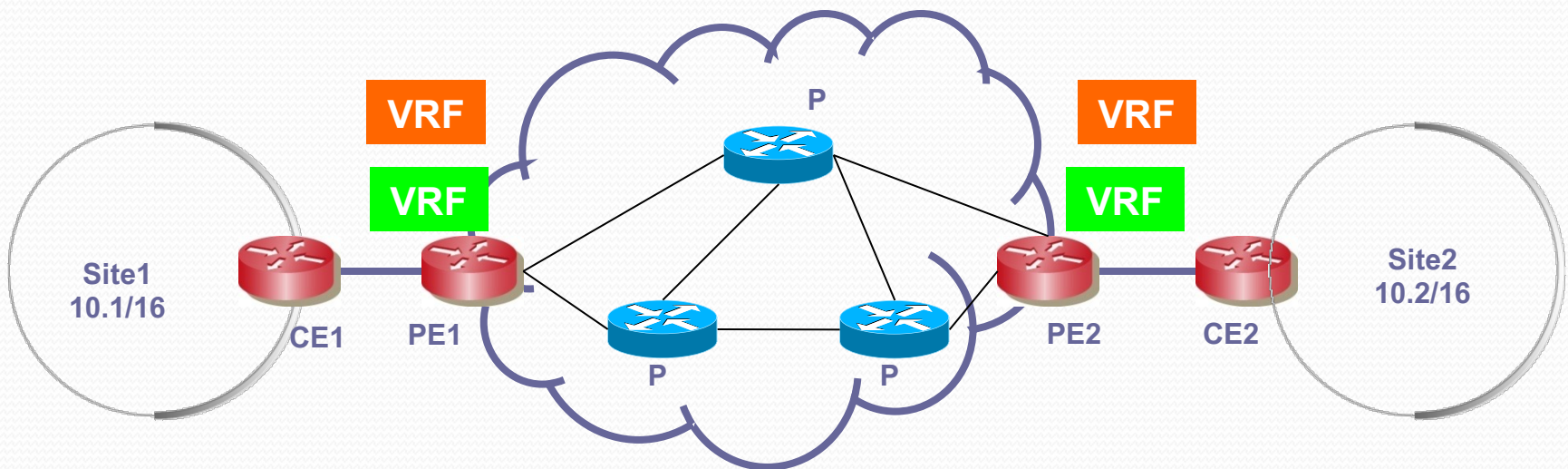
- Site 1 and Site 2 share the green VRF
- CE1 announces network 10.1/16 via E-BGP to PE1
- PE1 adds 10.1/16 to the green VRF using the RD identifier (Router Distinguished). PE1 determines that 10.1/16 has to be attached to the green VRF using the physical receiving port
- PE1 exports via I-BGP (using MBPG) the route defined by the VPN-IPv4@:
 - **Associates the route to the green VRF adding an extended communities (8B=RD)**
 - **Selects a MPLS label as “site-id” and adds it to the route** (e.g., label 353)
 - **Selects its loopback IP address as next-hop**



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● MPLS-VPN:

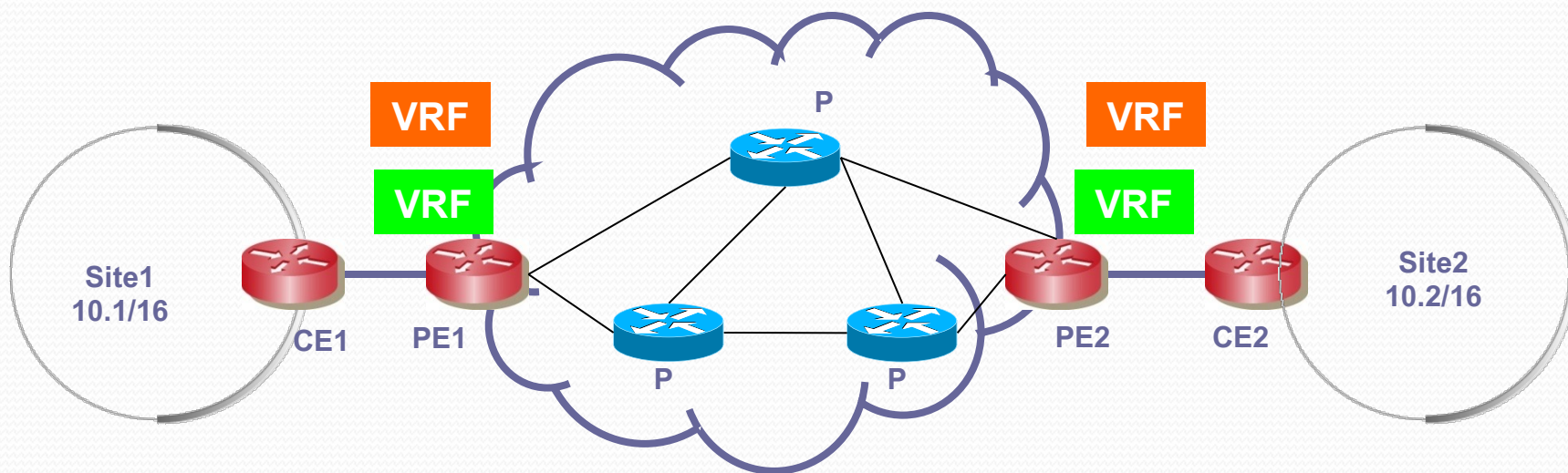
- **PE2 receives the VPN-IPv4@ route for 10.1/16 from PE1 and performs route-map filtering**
 - Extended BGP communities are inspected to determine if the route pertains to a known VRF
- PE2 accepts the route because it pertains to Green VRF
 - VRF determined based on extended community
 - PE2 records label 353 to use for forwarding customer packets to site 1
- PE2 also announces local Green VRF routing information to PE1



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● MPLS-VPN:

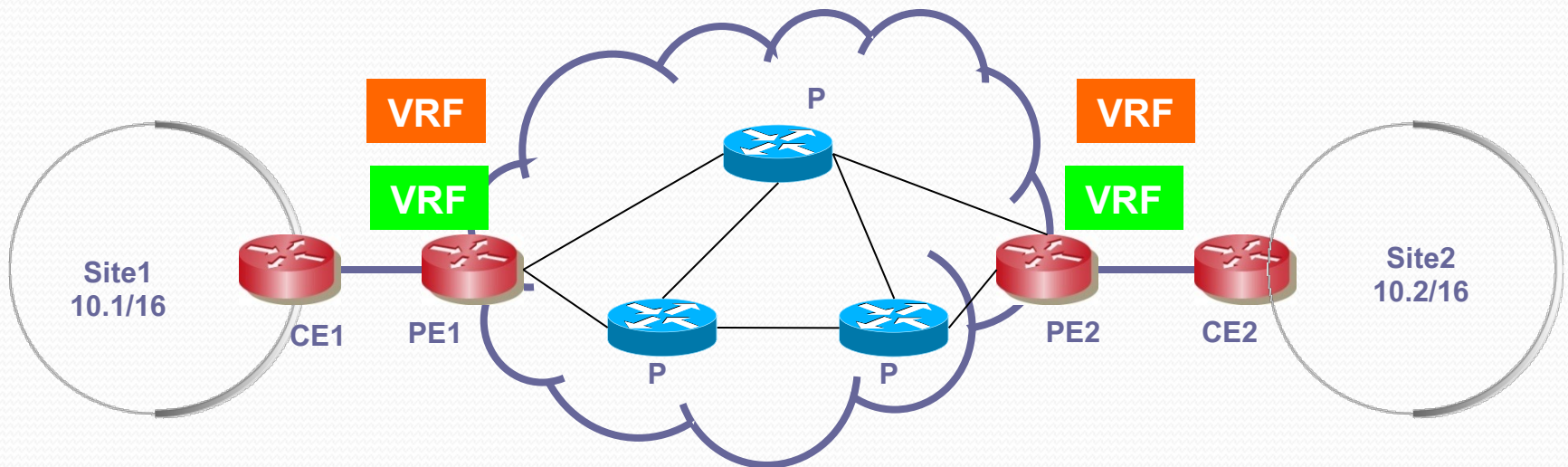
- Use MPLS to forward traffic. For that a MPLS route between PE1 and PE2 must be established
 - Created by PE2 when it learns the route announced by PE1 (and viceversa)
- E.g., label 979 propagated to all Ps in the path between PE1 and PE2
 - Different than "site label" 353
- There are two MPLS labels
 - **LSP tag:** Label 979 is used to forward packets in the MPLS network from PE1 to PE2
 - **Site-tag:** Label 353 is used to identify who is the Remote site (i.e., site 1)



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● MPLS-VPN:

- Host 10.2.1.1 in site2 communicates with host 10.1.1.1 in site1
- PE2 determines VRF based on receiving port, and looks up 10.1.1.1, obtaining:
 - MPLS Tag associated to remote site: 353
 - Next-hop of 10.1/16 route: PE1's loopback
 - MPLS Tag associated to reach PE1: 979
- "LSP tag" used to forward packets over the appropriate LSP
- "Site tag" used by remote PE to forward packets to the appropriate port



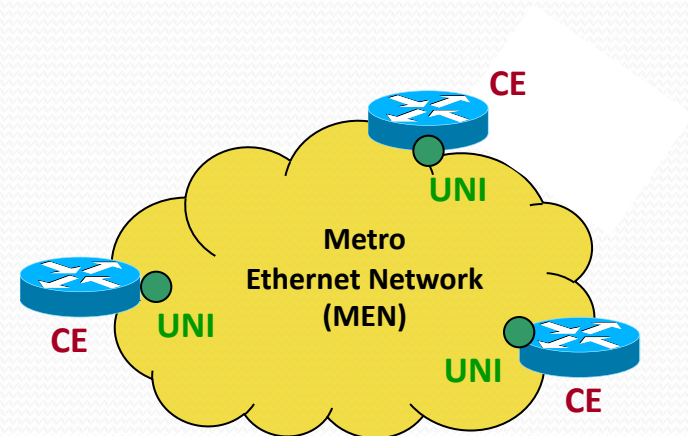
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● Metro Ethernet:

- **Goal:** Ethernet in the MAN (Metropolitan Area Network)
- **Interconnect Ethernet switches and routers with the operator or ISP to transport any service using broadband VPN (10/100/1000/10000 Mb/s)**
 - In the last years L2 VPN were over FR or ATM
 - Metro Ethernet may switch based in VLAN traffic (same way that FR does with the DLCI or ATM with the VC/VP)
- **Solutions**
 - Optical Ethernet or Native Ethernet
 - EFM Ethernet over the First Mile or IEEE802.3ah
 - Ethernet over MPLS (ISPs)
 - EoS: Ethernet over SDH
 - EoW: Ethernet over WDM (Wavelength Division Multiplexing)

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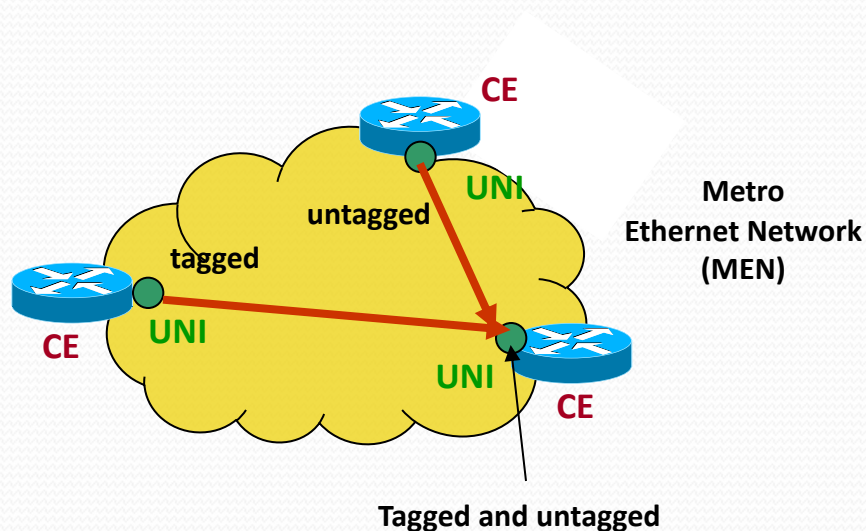
- When an ISP offers a Metro Ethernet service, **two logical entities called UNI** sited in **the CE transports frames through a logical channel called EVC**,
 - CE (Customer Equipment)**: switch (with IEEE 802.1Q – VLAN support) or router
 - UNI (User Network Interface)**: IEEE 802.3 PHY/MAC with 10/100/1000/10000 Mb/s and QoS support
 - MEN (Metro Ethernet Network)**: the operator can use any kind of L2 technology or L3 (FR, ATM, SONET/SDH, WDM, MPLS, etc)
 - EVC (Ethernet Virtual Connection)**: logical association between one or more UNI that transports frames from the origin UNI towards one (point-to-point) or more (point-to-multipoint) destination UNIs (allowing then the creation of a L2 VPN) → equivalent to a VLAN,
- Multiplexing services: a UNI associated to more than one EVC
- In Metro Ethernet there is **three types of services**:
 - Ethernet Line** (point-to-point)
 - Ethernet LAN** (multipoint-to-multipoint)
 - Ethernet Tree** (point-to-multipoint)



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• VLAN-tag support:

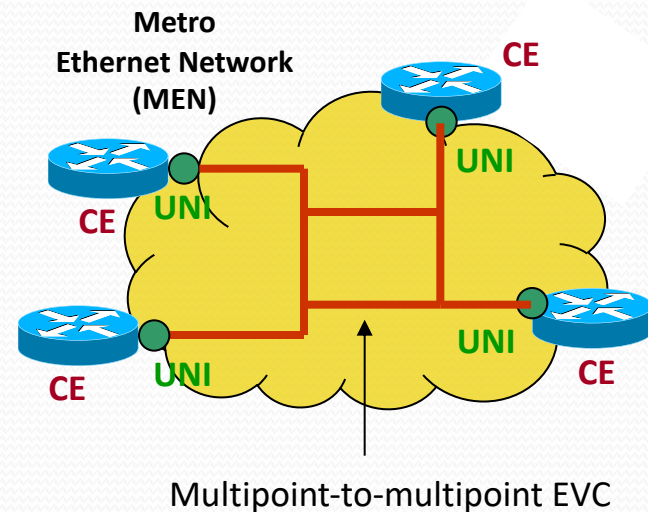
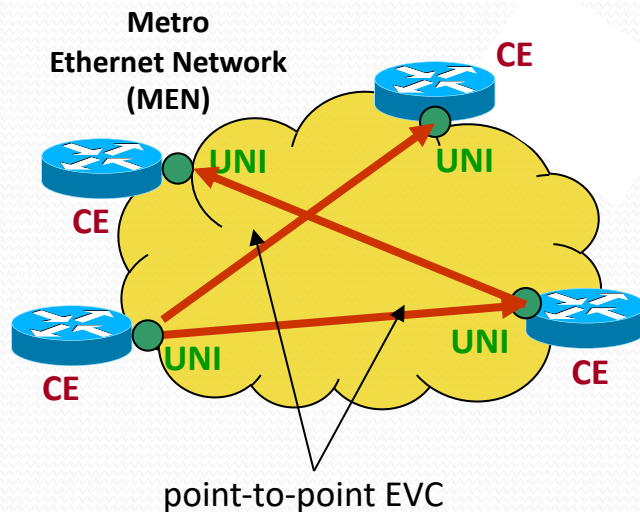
- The services can support only 802.1Q tagged frames, only not tagged frames or both of them
- The user may mark frames with a label and the provider may add a new one to tag the traffic of other users
 - CE-VLAN ID is the tagging performed by the user at the CE. This tagging is different to the provider's tagging



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• Metro Ethernet services:

- Point-to-point services (E-Line):
- Multipoint-to-multipoint services (E-LAN):



The **services** have attributes that represent them:

- **UNI attributes:** Tx media, data rate, VLAN support, multiplexing, ...
- **EVC attributes:** traffic parameters (CIR, PIR, EIR, CBS), QoS parameters (delay, jitter, losses), Service Class parameters (VLAN-ID), unicast delivery, multicast delivery, etc

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- **Ether-Line (E-Line) service**

- Can operate with both dedicated/switched bandwidth and is a **point-to-point architecture**,

- **EPL (Ethernet Private Line)**

- Is a point-to-point EVC where the user defines CIR, CBS, EIR, EBS, ...
- Can be seen as pure point-to-point where the EPL supports a **unique** EVC between two UNI's
- Since there is only one EVC, the user doesn't see the "VLAN tag"

- **EVPL (Ethernet Virtual Private Line)**

- Allows service multiplexing, thus, the point-to-point supports several EVC between two UNI's
- Exists a CIR and EIR and a metric for SLA's support
- Very similar to Frame Relay or ATM (where VC are multiplexed)
- Since there are several EVC, the user has to tag packets with a "VLAN tag" per EVC

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- **Ether-LAN (E-LAN) service**

- Can operate with both dedicated/switched bandwidth in a **multipoint-to-multipoint architecture**,

- **EPLan (Ethernet Private LAN)**

- Multipoint-to-multipoint connectivity between two or more UNI's
- Each UNI is only attached to one EVC (if the user wants other EVC he has to activate other UNI)
- Since there is only one EVC, the user will not see the "VLAN tag"

- **EVPLan (Ethernet Virtual Private LAN)**

- Same as VPLS (Virtual Private Lan Service), TLS (Transparent Lan Service) or VPSN (Virtual Private Switched Network)
- Multipoint-to-multipoint connectivity between two or more UNI's, with multiple EVC's support, the user has to tag packets with a "VLAN tag" per EVC

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- **Ether-Tree (E-Tree) service**

- **Point-to-multipoint connectivity** with dedicated bandwidth and tree topology
- It is not a any-to-any communication since there is a root UNI

